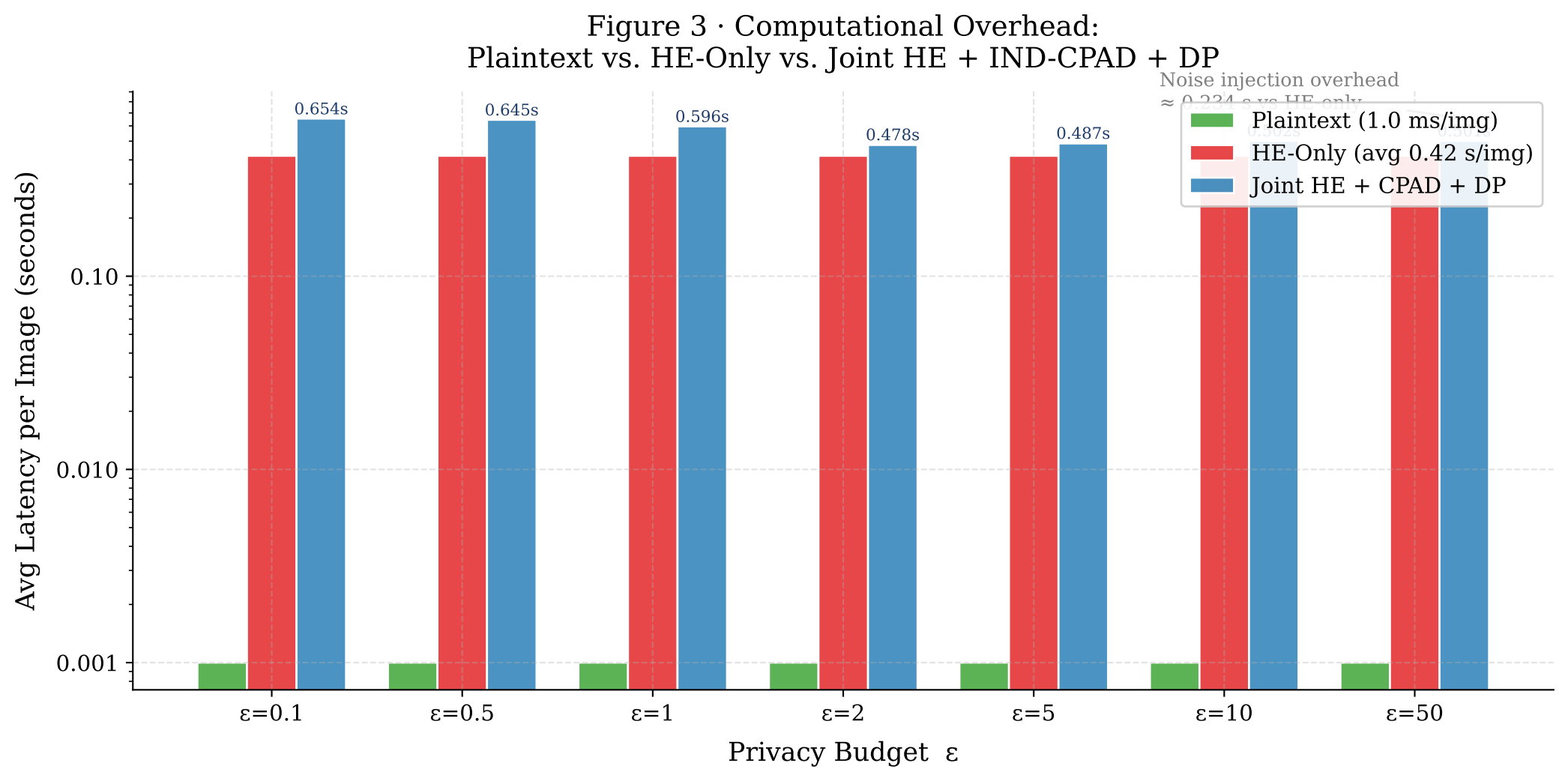
Figure 3 · Computational Overhead:
Plaintext vs. HE-Only vs. Joint HE + IND-CPAD + DP
Noise injection overhead
≈ 0.234 s vs HE-only
Plaintext (1.0 ms/img)
HE-Only (avg 0.42 s/img)
Joint HE + CPAD + DP
0.654s
0.645s
0.596s
0.478s
0.487s
Avg Latency per Image (seconds)
0.10
0.010
0.001
ε=0.1
ε=0.5
ε=1
ε=2
ε=5
ε=10
ε=50
Privacy Budget ε